

Shaan Patel

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Education

University of Texas at Arlington

Ph.D. and M.S. in Physics and Applied Physics, GPA: 4.00

2025

- Authored 6 peer-reviewed publications in leading astrophysics journals; presented research at 3 physics conferences

Georgia Institute of Technology

B.S. in Physics, Highest Honor

2019

Experience

Computational Astrophysics Researcher | UT Arlington

2021 – 2025

- Executed 50,000+ numerical Python simulations with the Exoplanets group, optimizing mathematical algorithms to reduce computation time and improve efficiency by 80–90%
- Processed and visualized multi-million-row time-series datasets to identify key dynamical patterns

Lead Undergraduate Researcher | Georgia Institute of Technology

2017 – 2019

- Led a research team in the Numerical Relativity Research Group simulating binary black hole dynamics on HPC clusters, analyzing high-dimensional gravitational wave data for LIGO research

Investment Analyst | Georgia Institute of Technology

2016 – 2017

- Conducted research, financial modeling, and valuation analyses for the Student Investments Committee to manage and pitch equity investments for a multi-million-dollar student-run endowment

Projects

AuroraNet: Capstone Aurora Prediction Project

- Engineered an end-to-end time-series pipeline in Python (pandas, NumPy) to clean 24 years of space weather data, training PyTorch LSTM models to achieve an RMSE of 0.642
- Deployed real-time inference endpoints via FastAPI and built an interactive Streamlit dashboard, containerizing the app with Docker and logging experiments using Weights & Biases (W&B)

LedgerLens: Deterministic Financial Analytics & Evidence-Grounded RAG

- Architected a Neuro-Symbolic RAG engine combining deterministic SEC XBRL analytics with hybrid semantic search (BM25, dense vector embeddings, cross-encoder reranking) and local LLMs (Ollama, Llama 3.2) enforcing Pydantic structured JSON outputs
- Engineered an offline evaluation framework achieving 71.4% Recall@5 and 0.548 MRR on a verified SEC Golden Dataset, deploying a batch-optimized FastAPI backend and interactive Streamlit UI decoupled from EDGAR ingestion I/O

SolarWindLakehouse: End-to-End PySpark, Delta Lake & MLFlow Pipeline

- Built a Medallion Lakehouse anomaly detection engine on Databricks using PySpark and Delta Lake to process space telemetry, using PySpark SQL windowing for feature engineering and MLflow to track, register, and serve a SparkML K-Means pipeline for batch predictions

Skills

Languages & Systems: Python, SQL, Bash, Git, Linux, PyTest, Pydantic

Cloud & Data Engineering: AWS, Databricks, Apache Spark/PySpark, Delta Lake, Unity Catalog

Generative AI & Information Retrieval: Retrieval-Augmented Generation (RAG), Local LLMs (Ollama), Hybrid Search (BM25 + Dense Vector Embeddings), Cross-Encoder Reranking, Sentence-Transformers

Machine Learning & MLOps: PyTorch, Scikit-learn, MLflow, Weights & Biases (W&B), Time Series

Data Science & Analytics: A/B Testing, Hypothesis Testing, Statistical Modeling, pandas, NumPy, SciPy, matplotlib, seaborn

Deployment & Backend: FastAPI, Streamlit, Docker, REST APIs